23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No : D9

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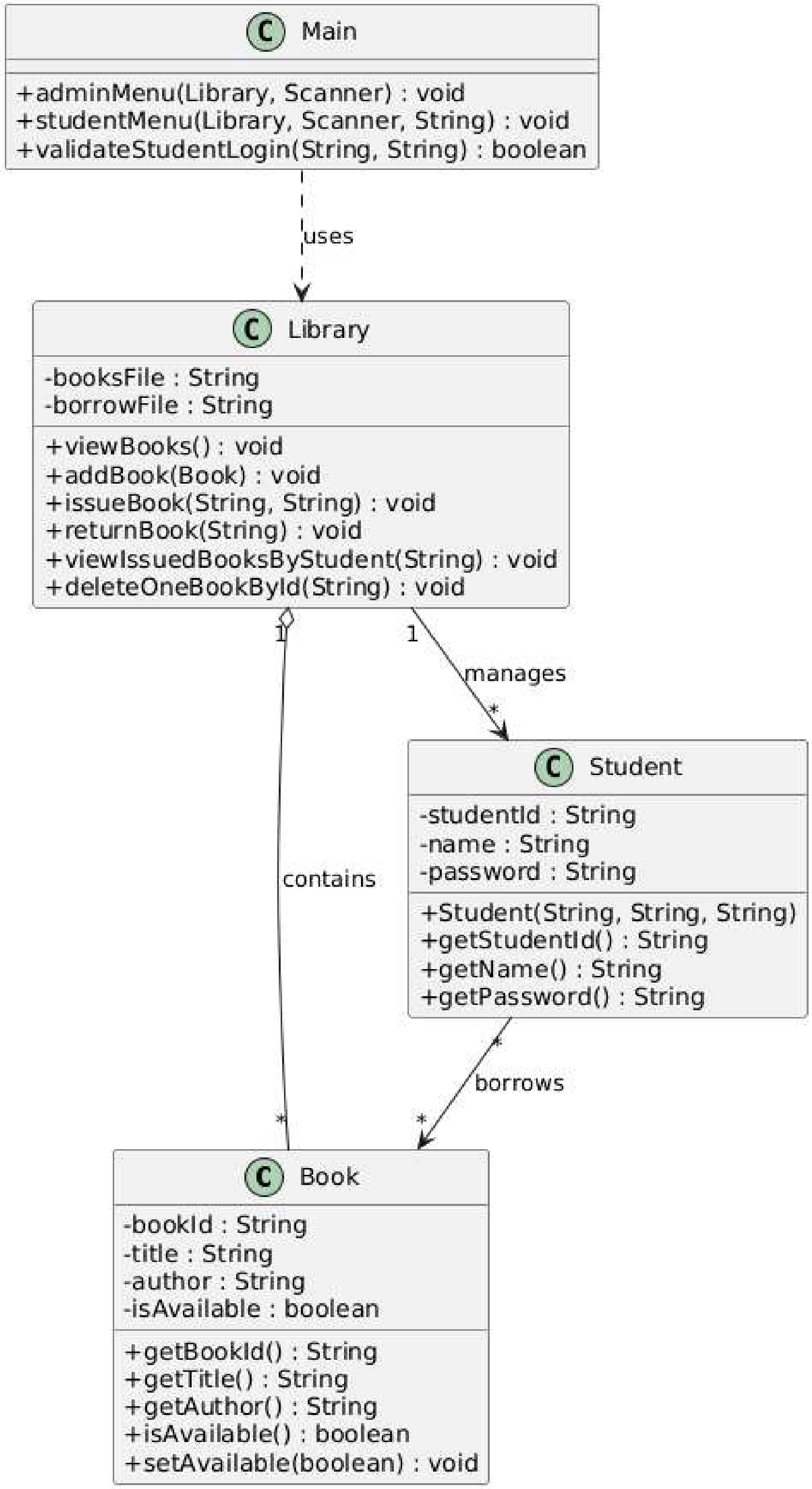
1. Problem Title : Library Management System

<u>STRUCTURAL DESIGN</u>

1. Class diagram



BEHAVIUORAL DESIGN

1. Use-case Diagram

a. List of Actors

1) ***Librarian*** – Manages the library system, including adding books, issuing and returning books, and maintaining records.

2) ***Student*** – Uses the system to view books, check availability, and view their borrowed books.

b. List of Use-cases

1. View Books:
Allows the student to view the list of available books in the library.

2. Manage User Info:
Allows the librarian to add, update, and maintain student information in the system.

3. Manage Book Info:
Allows the librarian to add new books, update existing book details, and maintain book records.

4. Check Book Availability:
Allows the system to verify whether a requested book is available for issuing.

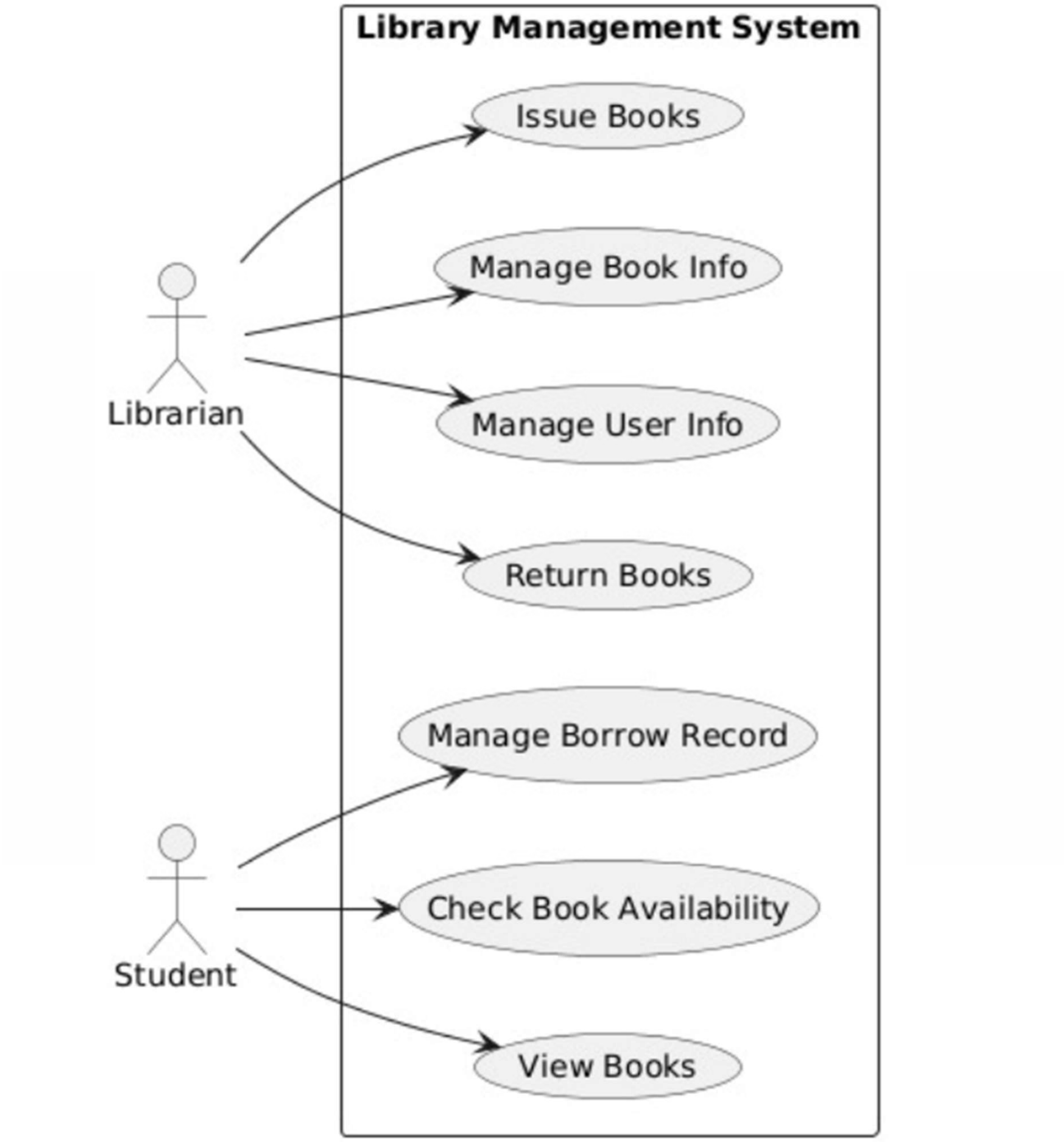
5. Issue Book:
Allows the librarian to issue a book to a student after verifying availability and student details.

6. Return Book:
Allows the librarian to process the return of a book and update its availability status.

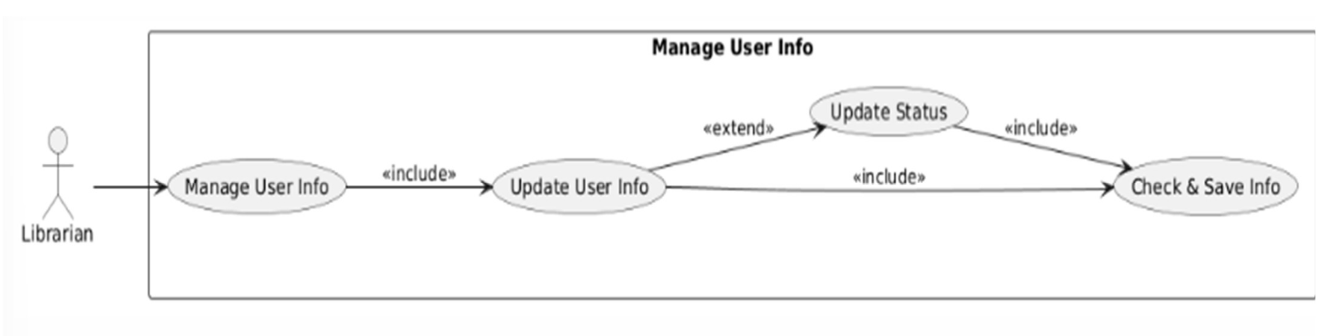
7. Manage Borrow Record:
Allows the system to maintain records of issued and returned books for tracking purposes.

c.Use-case Diagrams :

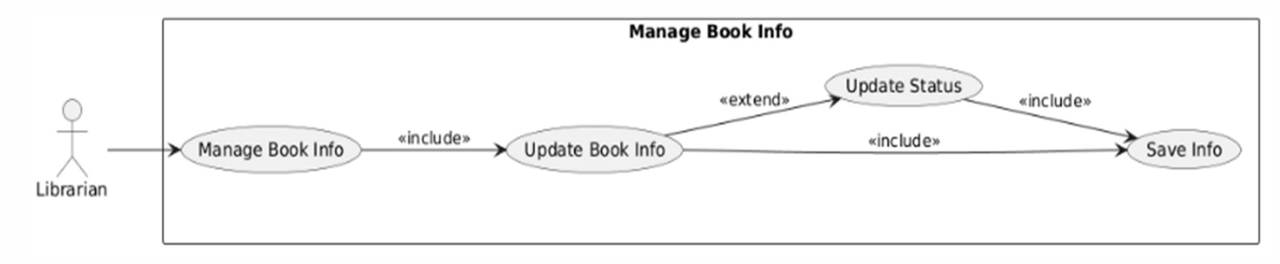
MAIN USE CASE DIAGRAM (SYSTEM LEVEL)



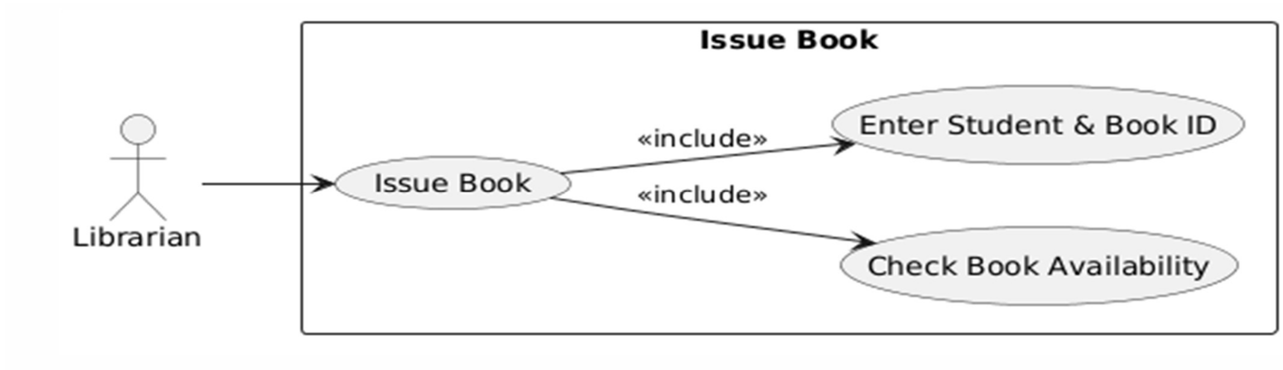
MANAGE USER INFO (DETAILED USE CASE)



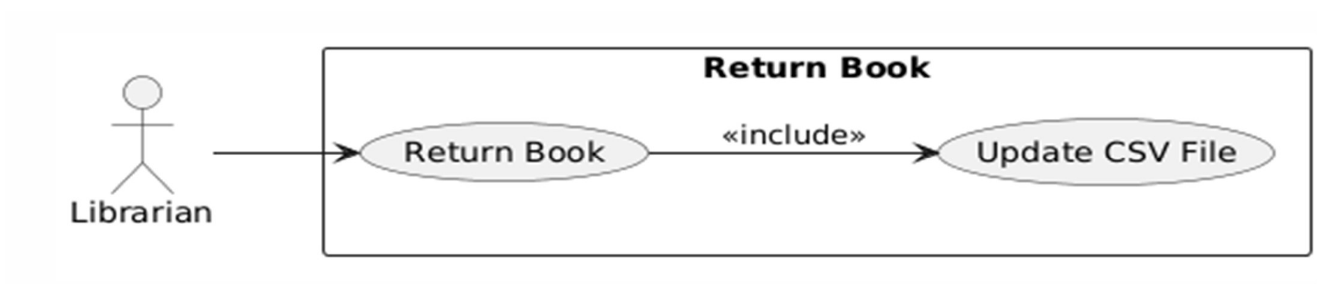
MANAGE BOOK INFO (DETAILED USE CASE)



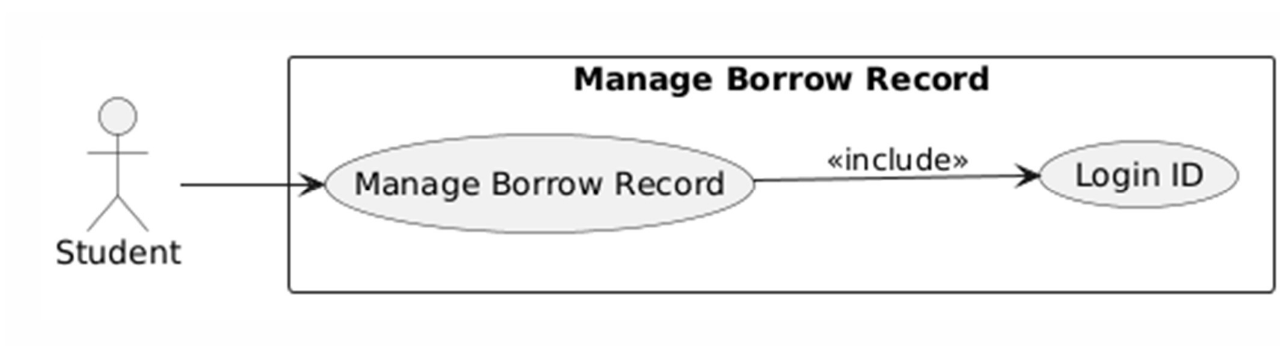
ISSUE BOOK (DETAILED USE CASE)



RETURN BOOK (DETAILED USE CASE)



MANAGE BORROW RECORD (DETAILED USE CASE)



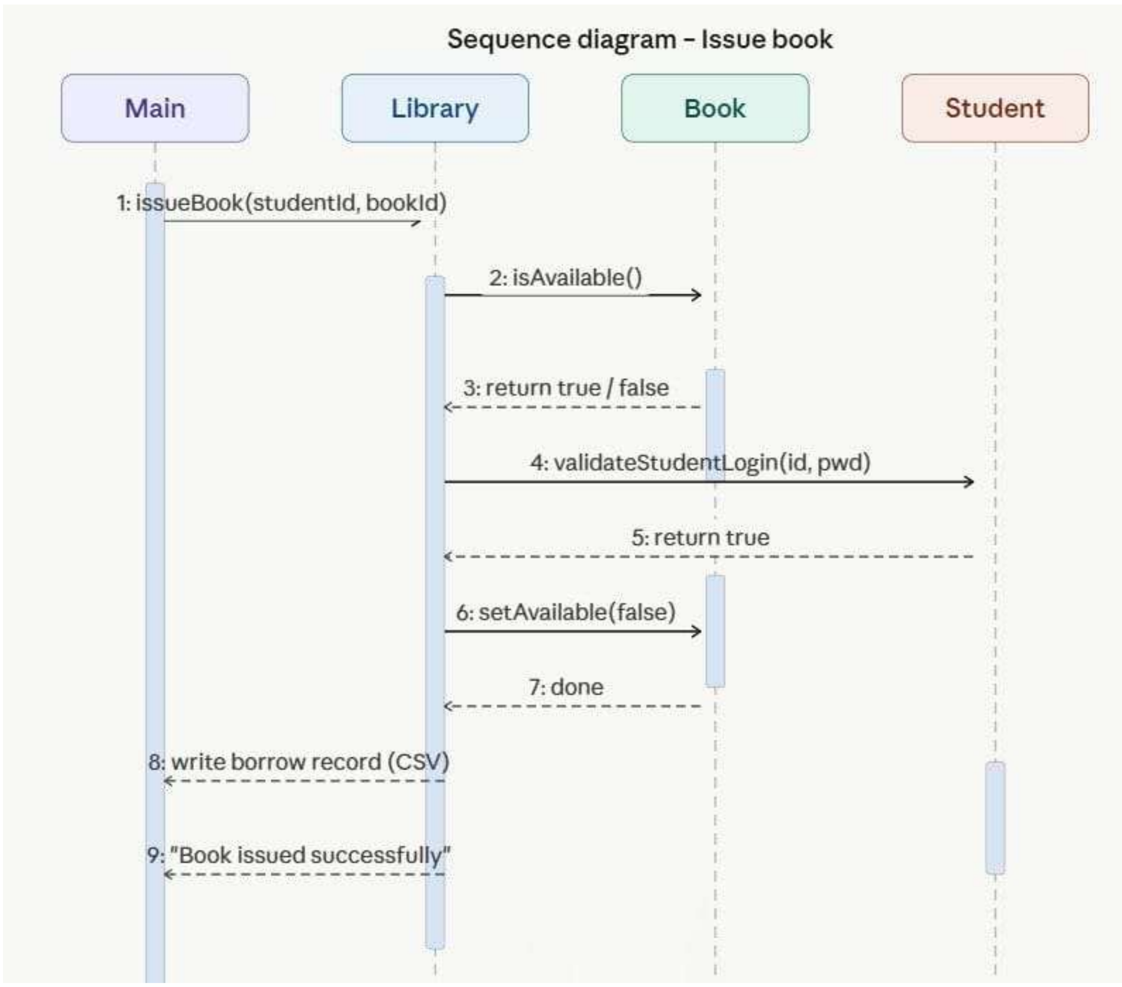
The use-case diagram represents the interaction between actors (Librarian and Student) and the system functionalities such as issuing, returning, and managing books.

2. Sequence diagram

a. Use Case: Issue Book

Scenario

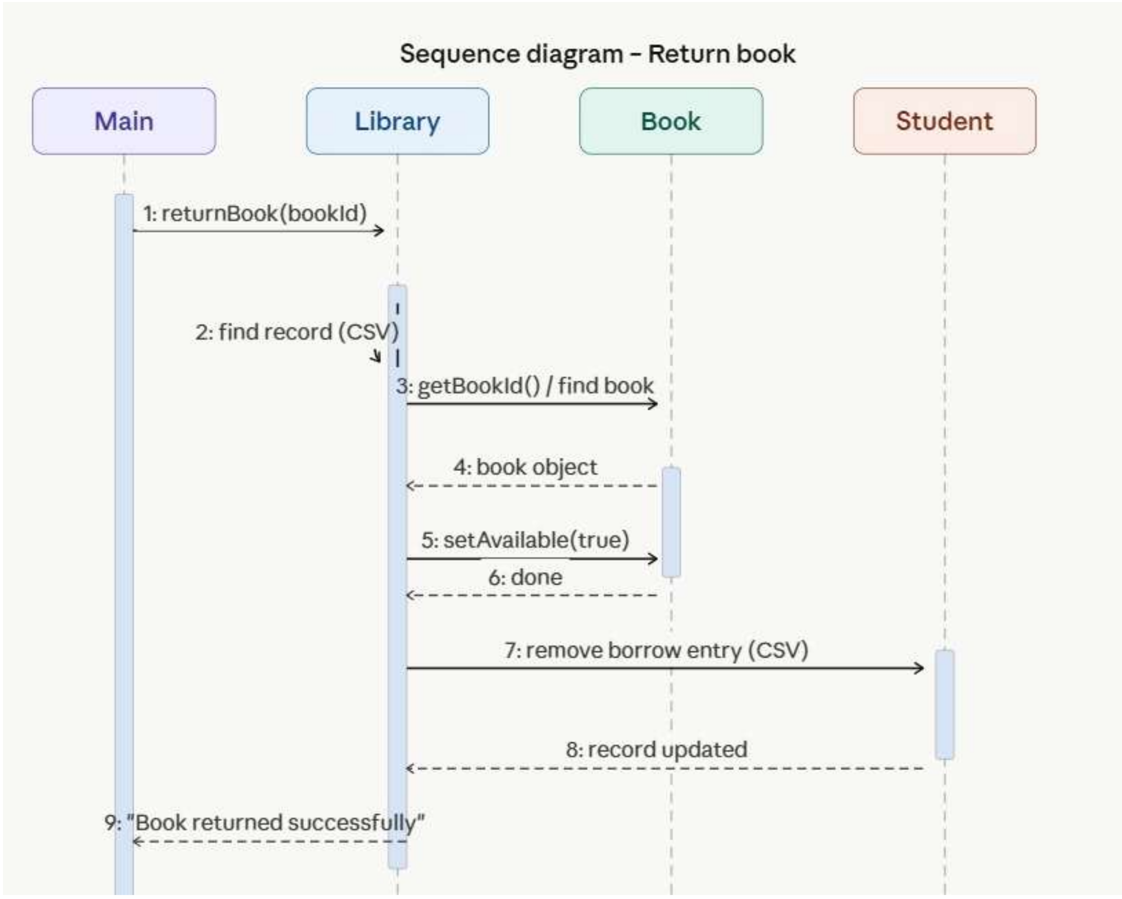
When a student requests a book, the librarian enters the student ID and book ID into the system. The system checks whether the requested book is available. If the book is available and the student credentials are valid, the system issues the book by marking it as unavailable. It then records the transaction in the borrow file (CSV) and confirms that the book has been issued successfully.



b. Use Case: Return Book

Scenario

When a student returns a book, the librarian enters the book ID into the system. The system locates the corresponding borrow record and identifies the book. It updates the book's availability status to available and removes the borrow entry from the records. Finally, the system confirms that the book has been returned successfully.


Sequence diagram - Return book
Main
Library
Book
Student
1: returnBook(bookId)
2: find record (CSV)
3: getBookId() / find book
4: book object
5: setAvailable(true)
6: done
7: remove borrow entry (CSV)
8: record updated
9: "Book returned successfully"